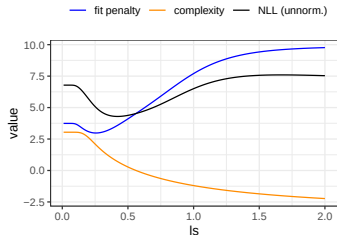
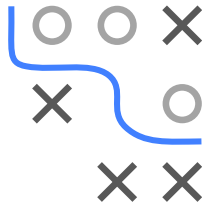


Introduction to Machine Learning

Gaussian Processes

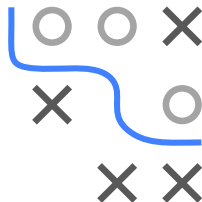
Training of a Gaussian Process



Learning goals

- Training of GPs via Maximum Likelihood estimation of its hyperparameters
- Computational complexity is governed by matrix inversion of the covariance matrix

TRAINING OF A GAUSSIAN PROCESS



- All we need for GP predictions (in regression): matrix computations
- Implicit assumption: fully specified cov function, incl. hyperparams
- Nice GP property: numerical hyperparams of given cov function can be learned during training

TRAINING VIA MARGINAL LIKELIHOOD

- Let

$$y = f(\mathbf{x}) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

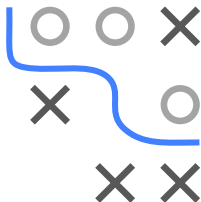
with $f \sim \mathcal{GP}(\mathbf{0}, k(\cdot, \cdot | \theta))$ for hyperparam config θ

- This yields $\mathbf{y} \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_y)$ with $\mathbf{K}_y = \mathbf{K} + \sigma^2 \mathbf{I}$

- We get the **negative marginal log-likelihood** / evidence

$$\begin{aligned} -\log p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) &= -\log \left[(2\pi)^{-n/2} |\mathbf{K}_y|^{-1/2} \exp\left(-\frac{1}{2} \mathbf{y}^T \mathbf{K}_y^{-1} \mathbf{y}\right) \right] \\ &= \frac{1}{2} \mathbf{y}^T \mathbf{K}_y^{-1} \mathbf{y} + \frac{1}{2} \log |\mathbf{K}_y| + \frac{n}{2} \log 2\pi \end{aligned}$$

- $-\log p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta})$ depends on $\boldsymbol{\theta}$ via $\mathbf{K}_y \Rightarrow$ optimize for $\boldsymbol{\theta}$
- GP training optimizes kernel hyperparams by minimizing the neg. marginal likelihood, balancing data fit ($\frac{1}{2}\mathbf{y}^\top \mathbf{K}_y^{-1} \mathbf{y}$) and model complexity ($\frac{1}{2} \log |\mathbf{K}_y|$)

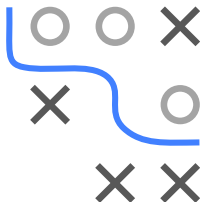


NEGATIVE LOG-LIKELIHOOD COMPONENTS

- Consider common cov type parameterized by length-scale: $\theta = \ell$ (for simplicity: assume univariate rather than different for each dim)
- E.g., squared exponential kernel

$$k(\mathbf{x}, \tilde{\mathbf{x}}) = \exp\left(-\frac{1}{2\ell^2} \|\mathbf{x} - \tilde{\mathbf{x}}\|^2\right)$$

- For small ℓ , cov decays quickly $\Rightarrow$ local model, $\mathbf{K}_y$ approaches $\sigma^2 \mathbf{I}$
- **Data fit:** small $\ell \Rightarrow$ small $\mathbf{K}_y^{-1} \mathbf{y}$ (cov structure aligns well with observed data; small residuals) $\Rightarrow$ small $\frac{1}{2} \mathbf{y}^T \mathbf{K}_y^{-1} \mathbf{y}$
- **Complexity penalty:** small $\ell \Rightarrow$ large $\frac{1}{2} \log |\mathbf{K}_y|$
- Normalization constant $\frac{n}{2} \log 2\pi$

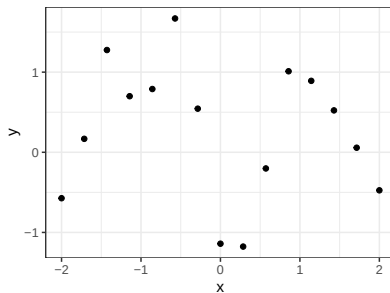
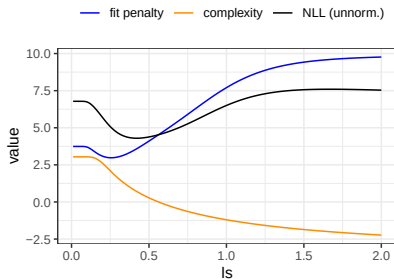
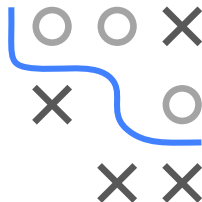


NLL COMPONENTS: EXAMPLE

- Let $f \sim \mathcal{GP}(\mathbf{0}, k(\cdot, \cdot))$ with squared exp kernel

$$k(\mathbf{x}, \tilde{\mathbf{x}}) = \exp\left(-\frac{1}{2\ell^2} \|\mathbf{x} - \tilde{\mathbf{x}}\|^2\right)$$

- Minimize NLL by trading off data fit & complexity

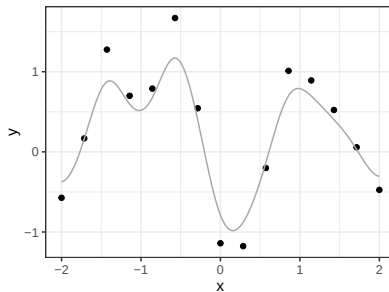
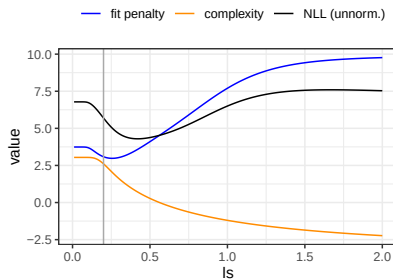
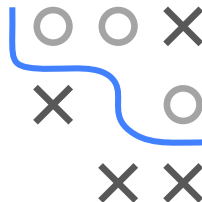


NLL COMPONENTS: EXAMPLE

- Let $f \sim \mathcal{GP}(\mathbf{0}, k(\cdot, \cdot))$ with squared exp kernel

$$k(\mathbf{x}, \tilde{\mathbf{x}}) = \exp\left(-\frac{1}{2\ell^2} \|\mathbf{x} - \tilde{\mathbf{x}}\|^2\right)$$

- Minimize NLL by trading off data fit & complexity
- $\ell = 0.2$: good fit but high complexity

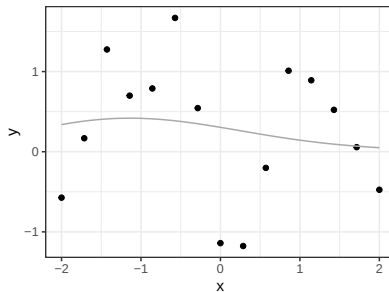
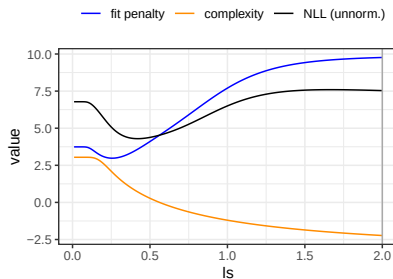
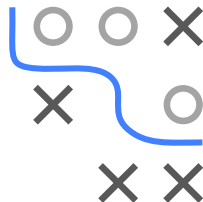


NLL COMPONENTS: EXAMPLE

- Let $f \sim \mathcal{GP}(\mathbf{0}, k(\cdot, \cdot))$ with squared exp kernel

$$k(\mathbf{x}, \tilde{\mathbf{x}}) = \exp\left(-\frac{1}{2\ell^2} \|\mathbf{x} - \tilde{\mathbf{x}}\|^2\right)$$

- Minimize NLL by trading off data fit & complexity
- $\ell = 2$: smooth but poor fit

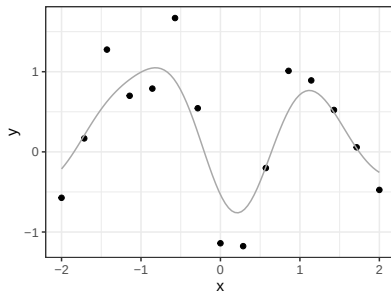
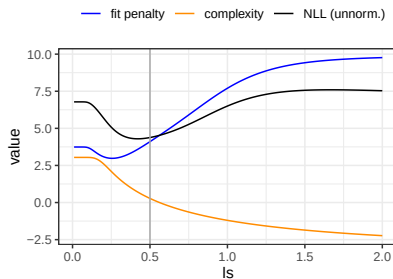
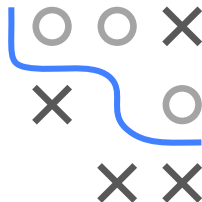


NLL COMPONENTS: EXAMPLE

- Let $f \sim \mathcal{GP}(\mathbf{0}, k(\cdot, \cdot))$ with squared exp kernel

$$k(\mathbf{x}, \tilde{\mathbf{x}}) = \exp\left(-\frac{1}{2\ell^2} \|\mathbf{x} - \tilde{\mathbf{x}}\|^2\right)$$

- Minimize NLL by trading off data fit & complexity
- $\ell = 0.5$: balancing data fit and smoothness



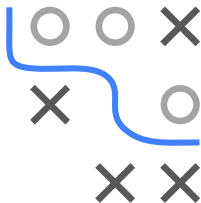
OPTIMIZING KERNEL HYPERPARAMETERS

- Set partial derivatives wrt hyperparams to 0

$$\begin{aligned}\frac{\partial}{\partial \theta_j} \log p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) &= \frac{\partial}{\partial \theta_j} \left(-\frac{1}{2} \mathbf{y}^T \mathbf{K}_y^{-1} \mathbf{y} - \frac{1}{2} \log |\mathbf{K}_y| - \frac{n}{2} \log 2\pi \right) \\ &= \frac{1}{2} \mathbf{y}^T \mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial \theta_j} \mathbf{K}^{-1} \mathbf{y} - \frac{1}{2} \text{tr} \left(\mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial \theta_j} \right) \\ &= \frac{1}{2} \text{tr} \left((\mathbf{K}^{-1} \mathbf{y} \mathbf{y}^T \mathbf{K}^{-1} - \mathbf{K}^{-1}) \frac{\partial \mathbf{K}}{\partial \theta_j} \right)\end{aligned}$$

using $\frac{\partial}{\partial \theta_j} \mathbf{K}^{-1} = -\mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial \theta_j} \mathbf{K}^{-1}$ and $\frac{\partial}{\partial \theta} \log |\mathbf{K}| = \text{tr} \left(\mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial \theta} \right)$

- Bottleneck: inverting (or rather, decomposing) $\mathbf{K}$
 $\Rightarrow \mathcal{O}(n^3)$ for standard methods
- Only $\mathcal{O}(n^2)$ per hyperparam / partial derivative once $\mathbf{K}^{-1}$ is known
 $\Rightarrow$ small overhead, so use gradient-based optim



STRATEGIES FOR BIG DATA

- Kernels that yield sparse $\mathbf{K} \Rightarrow$ cheaper to invert
- Subsample data $\Rightarrow \mathcal{O}(m^3)$ with $m^3 \ll n^3$
- **Bayesian committee**: combine estimates on different size- m estimates $\Rightarrow \mathcal{O}(nm^2)$
- **Nystrom approx**: low-rank approx from representative subset (“inducing points”): $\mathbf{K} \approx \mathbf{K}_{nm}\mathbf{K}_{mm}^{-1}\mathbf{K}_{mn} \Rightarrow \mathcal{O}(nmk + m^3)$ for rank- k -approx inverse of $\mathbf{K}_{mm}$
- Exploit structure in $\mathbf{K}$ induced by kernels $\Rightarrow$ exact but complicated solutions; kernel-specific
- Still active research area

